

PAPER_385 — Canonical 7-System UQFF Parameter Registry

Source: grok_share_11254865.txt, lines ~6700–6850 (confirmed 9400–10322 in main())

Section: MUGESystem struct initializations for all 7 canonical validation systems

Session: 104 (Complete Re-Analysis — full 18-field per-system registry not formalized in prior papers)

CP4 Class: Canonical7SystemUQFFParameterRegistryCalculator (CP4 #36)

Abstract

$$F_{U,Bi} = \kappa \cdot \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + U_{bi})$$

This paper presents a UQFF analysis of Canonical 7-System UQFF Parameter Registry, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_379 compared the compressed vs resonance MUGE outputs for all 7 canonical systems. PAPER_377 noted the 18-field CSV format. But no paper explicitly documented all **18 parameters for each of the 7 systems** as a canonical reference registry.

This paper establishes that registry — the authoritative configuration table for UQFF validation that all C++ and Python implementations must match.

2. The 18-Field MUGESystem Parameter Set

The MUGESystem struct defines 18 fields for each astrophysical system:

Field	Symbol	Description	Units
1. name	—	System identifier	string
2. I	I	Electric current	A
3. A	A	Cross-sectional area	m ²
4. omega1	ω_1	Upper frequency	rad/s
5. omega2	ω_2	Lower frequency	rad/s
6. Vsys	V_sys	System volume	m ³
7. vexp	v_exp	Expansion velocity	m/s
8. t	t	System age	s
9. z	z	Redshift	—
10. ffluid	f_fluid	Fluid oscillation frequency	Hz
11. M	M	Total mass	kg
12. r	r	Characteristic radius	m
13. B	B	Magnetic field	T
14. Bcrit	B_crit	Critical magnetic field	T
15. rho_fluid	ρ_f	Fluid density	kg/m ³
16. g_local	g_local	Local gravitational surface acc.	m/s ²
17. M_DM	M_DM	Dark matter mass	kg
18. delta_rho_rho	$\delta\rho/\rho$	Fractional density contrast	—

3. Canonical Parameter Registry — All 7 Systems

System 1: SGR1745 — Magnetar SGR 1745-2900

Field	Value
I	1×10^{21} A
A	3.142×10^8 m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	4.189×10^{12} m ³
v_exp	1×10^3 m/s
t	3.799×10^{10} s
z	0.0009
f_fluid	1.269×10^{-14} Hz
M	2.984×10^{30} kg
r	1×10^4 m
B	1×10^{10} T
B_crit	1×10^{11} T
ρ_f	1×10^{15} kg/m ³
g_local	1.991×10^{12} m/s ²
M_DM	1×10^{28} kg
$\delta\rho/\rho$	0.1

System 2: Sag A* — Sagittarius A* (SMBH)

Field	Value
I	1×10^{23} A
A	2.813×10^{30} m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	3.552×10^{45} m ³
v_exp	5×10^6 m/s
t	3.786×10^{14} s
z	0.0009
f_fluid	3.465×10^{-8} Hz
M	8.155×10^{36} kg
r	1×10^{12} m
B	1×10^{-5} T
B_crit	1×10^{-4} T
ρ_f	1×10^{-19} kg/m ³
g_local	5.443×10^2 m/s ²
M_DM	1×10^{38} kg
$\delta\rho/\rho$	0.01

System 3: Tapestry — Tapestry of Blazing Starbirth

Field	Value
I	1×10^{22} A
A	1×10^{35} m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	1×10^{53} m ³
v_exp	1×10^4 m/s
t	3.156×10^{13} s
z	0.0
f_fluid	1×10^{-12} Hz
M	1.989×10^{35} kg
r	3.086×10^{17} m
B	1×10^{-9} T
B_crit	1×10^{-8} T
ρ_f	1×10^{-21} kg/m ³
g_local	1.39×10^{-15} m/s ²
M_DM	1×10^{36} kg
$\delta\rho/\rho$	0.01

System 4: Westerlund 2 — Westerlund 2 Star Cluster

Field	Value
I	1×10^{22} A
A	1×10^{35} m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	1×10^{53} m ³
v_exp	1×10^4 m/s
t	3.156×10^{13} s
z	0.0
f_fluid	1×10^{-12} Hz
M	1.989×10^{35} kg
r	3.086×10^{17} m
B	1×10^{-9} T
B_crit	1×10^{-8} T
ρ_f	1×10^{-21} kg/m ³
g_local	1.39×10^{-15} m/s ²
M_DM	1×10^{36} kg
$\delta\rho/\rho$	0.01

Note: Tapestry and Westerlund 2 share identical parameters — they represent two systems in the same star-forming complex with equivalent physical scale.

System 5: Pillars of Creation

Field	Value
I	1×10^{21} A
A	2.813×10^{32} m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	3.552×10^{48} m ³
v_exp	2×10^3 m/s
t	3.156×10^{13} s
z	0.0
f_fluid	8.457×10^{-14} Hz
M	1.989×10^{32} kg
r	9.46×10^{15} m
B	1×10^{-10} T
B_crit	1×10^{-9} T
ρ_f	1×10^{-23} kg/m ³
g_local	2.979×10^{-10} m/s ²
M_DM	1×10^{32} kg
$\delta\rho/\rho$	0.05

System 6: Rings of Relativity — Rings of Relativity Gravitational Lens

Field	Value
I	1×10^{22} A
A	1×10^{35} m ²
ω_1	1×10^{12} rad/s
ω_2	9.99×10^{11} rad/s
V_sys	1×10^{54} m ³
v_exp	1×10^5 m/s
t	3.156×10^{14} s
z	0.01
f_fluid	1×10^{-9} Hz
M	1.989×10^{36} kg
r	3.086×10^{17} m
B	1×10^{-10} T
B_crit	1×10^{-9} T
ρ_f	1×10^{-28} kg/m ³
g_local	1.391×10^{-14} m/s ²
M_DM	1×10^{38} kg
$\delta\rho/\rho$	0.02

System 7: Student's Guide — Student's Guide to the Universe (Cosmological)

Field	Value
I	1×10^{24} A

Field	Value
A	$1 \times 10^{52} \text{ m}^2$
ω_1	$1 \times 10^{12} \text{ rad/s}$
ω_2	$9.99 \times 10^{11} \text{ rad/s}$
V_sys	$1 \times 10^{80} \text{ m}^3$
v_exp	$3 \times 10^8 \text{ m/s}$
t	$4.35 \times 10^{17} \text{ s}$
z	0.0
f_fluid	$1 \times 10^{-18} \text{ Hz}$
M	$1 \times 10^{53} \text{ kg}$
r	$1 \times 10^{26} \text{ m}$
B	$1 \times 10^{-15} \text{ T}$
B_crit	$1 \times 10^{-14} \text{ T}$
ρ_f	$1 \times 10^{-26} \text{ kg/m}^3$
g_local	$6.67 \times 10^{-33} \text{ m/s}^2$
M_DM	$1 \times 10^{53} \text{ kg}$
$\delta\rho/\rho$	0.001

4. CSV Format (18-Field Standard)

The canonical CSV header for UQFF system configuration files:

```
name,I,A,omega1,omega2,Vsys,vexp,t,z,ffluid,M,r,B,Bcrit,rho_fluid,g_local,M_DM,delta_rho
sgr1745,1e21,3.142e8,1e12,9.99e11,4.189e12,1e3,3.799e10,0.0009,1.269e-14,2.984e30,1e4,1
sagA,1e23,2.813e30,1e12,9.99e11,3.552e45,5e6,3.786e14,0.0009,3.465e-8,8.155e36,1e12,1e-
5,1e-4,1e-19,5.443e2,1e38,0.01
tapestry,1e22,1e35,1e12,9.99e11,1e53,1e4,3.156e13,0.0,1e-12,1.989e35,3.086e17,1e-
9,1e-8,1e-21,1.39e-15,1e36,0.01
westerlund,1e22,1e35,1e12,9.99e11,1e53,1e4,3.156e13,0.0,1e-12,1.989e35,3.086e17,1e-
9,1e-8,1e-21,1.39e-15,1e36,0.01
pillars,1e21,2.813e32,1e12,9.99e11,3.552e48,2e3,3.156e13,0.0,8.457e-14,1.989e32,9.46e15
10,1e-9,1e-23,2.979e-10,1e32,0.05
rings,1e22,1e35,1e12,9.99e11,1e54,1e5,3.156e14,0.01,1e-9,1.989e36,3.086e17,1e-
10,1e-9,1e-28,1.391e-14,1e38,0.02
student_guide,1e24,1e52,1e12,9.99e11,1e80,3e8,4.35e17,0.0,1e-18,1e53,1e26,1e-
15,1e-14,1e-26,6.67e-33,1e53,0.001
```

5. System Scale Spectrum

System	Class	r (m)	M (kg)	Physics Domain
SGR1745	Magnetar	1×10^4	2.984×10^{30}	NS surface gravity
Sag A*	SMBH	1×10^{12}	8.155×10^{36}	Galactic center
Pillars	GMC	9.46×10^{15}	1.989×10^{32}	Stellar nursery
Tapestry	LMC-scale	3.086×10^{17}	1.989×10^{35}	Star-forming complex
Westerlund 2	Cluster	3.086×10^{17}	1.989×10^{35}	OB star cluster
Rings	Lens	3.086×10^{17}	1.989×10^{36}	Gravitational lens
Student's Guide	Cosmological	1×10^{26}	1×10^{53}	Observable universe

The 7 systems span **22 orders of magnitude** in radius (10^4 m to 10^{26} m) and **23 orders in mass** (10^{30} kg to 10^{53} kg), making this the most comprehensive UQFF multi-scale validation suite in the codebase.

6. Canonical Computed Results Summary

For reference — results from PAPER_379 (full comparison) and PAPER_382/384 (per-term):

System	Compressed MUGE (m/s^2)	Resonance MUGE (m/s^2)	Ratio
SGR1745	1.782×10^{39}	1.773×10^{-9}	10^{48}
Sag A*	2.966×10^{34}	4.105×10^{29}	$\sim 10^5$
Tapestry	$\sim \text{GM}/r^2$	fluid-dominated	converge
Westerlund 2	$\sim \text{GM}/r^2$	fluid-dominated	converge
Pillars	$\sim \text{GM}/r^2$	fluid-dominated	converge
Rings	$\sim \text{GM}/r^2$	fluid-dominated	converge
Student's Guide	$\sim \text{GM}/r^2$	fluid-dominated	converge

7. References Within Codebase

- PAPER_371: Resonance MUGE framework — 12-term formula
- PAPER_372: Compressed MUGE framework — 8-term formula
- PAPER_379: Dual-model 7-system comparison table
- PAPER_377: `load_muge_systems()` CSV parser (18-field format)
- `WormholeMUGETermImplSafetyCalculator` (CP4 #26): 7-system dict
- `MUGESuperconductive12TermResonanceCalculator` (CP4 #14): uses `sagA_dataset`

Source: `grok_share_11254865.txt` lines ~6700–6850 + lines ~9400–10322 (`main()` C++ impl.) | Session 104 | First formal 18-field canonical parameter registry for all 7 UQFF validation systems